26.08.14

New story:

Gentle manipulation of objects are studied with imitation learning approaches, where the demo comes from human teleoperating real robots. [[TFGripper, 2026]]

However, gentle manipulation of real world objects is a precise manipulation with many variations (like pose, size, shape, stiffness, etc.). A 100 or so demonstration hardly covers enough conditions and does not perform robustly under real-world setups (maybe show examples). Plus precise manipulation requires significant amount of demo to reach performance requirement [[curse-of-precision, 2026]]

Sim2real is a promising direction as it allows an autonomous expert to collect diverse experience to cover the conditions. We study the applicability of this to gentle manipulation of real world food.

Initial story:

Gentle object handling has predominantly been studied from a control perspective through the lens of tactile sensing [[tactile-dp, 2025], [VT-WAM, 2026]], which provides rich, direct feedback. Consequently, the majority of existing frameworks necessitate the active presence of tactile sensors during task execution. However, physical tactile sensors are not always viable due to spatial, financial, or mechanical constraints. A natural question then emerges: is it possible to achieve gentle object manipulation without relying on tactile sensors at all, while maintaining performance comparable to tactile-rich systems? Addressing this question also enhances robotic resilience; deploying a tactile-free backup controller can mitigate object damage in the event of sensor failure, providing a layer of analytical redundancy and fault tolerance [yifei's survey]. While some existing works leverage tactile information exclusively during training [curiosity-gentle-manip, HapticVLA, FD-VLA, VITaL], for example via cross-modal distillation [[v2t, 2025], [TacImag, 2026], [FELT, 2026]], they still rely on physical sensors to collect real-world training data—a process that is often tedious and limits scalability. Conversely, we investigate whether physical tactile sensors can be bypassed entirely across the whole pipeline—including training and deployment. To this end, we propose a sim-to-real pipeline that leverages privileged simulation-only data (e.g., object internal stress) in a deformable physics engine. This privileged information is used to train a student policy conditioned solely on point clouds and proprioceptive signals, which is then deployed zero-shot in the real world.

Some potential issues with the argument above:

(1) If tactiles are not available due to some constraints, would a depth camera also be unavailable for the same reason?
(2) Why not use tactile sensors in simulation but stress?

    (a) It is possible to use tactiles, and using it fits our motivation. However, stress provides direct information for object damage.
    (b) Tactile can be observed as an observation of stress, while stress is the most direct quantity
(3) What about gentle manipulation of rigid and/or articulated objects? Are these tasks applicable by the proposed method?
(4) We say distillation approaches are tedious, but a reviewer may attack us saying sim2real is tedious too.

Wrote in 2026.07.01:

A potential challenge is the variations of size and property of real-world deformable and fragile objects. It is questionable if a DP3 policy can adapt to the size of the objects - this needs primary testing on recoverable soft objects like sponge.

Another strong argument for a sim2real approach is the possibility to encounter a diverse range of scenarios. Where real-world imitation learning policies are inherently limited by the number of demonstrations. In-sim RL allows minimal human effort (compared to real-world and in-sim imitation learning) AND allows different scenarios to be encountered. “Reinforcement learning (RL) can overcome the demonstration-coverage limitation of imitation learning (IL) by allowing robots to improve through trial-and-error interaction beyond the states observed in demonstrations.” [WorldSample, 2026]

Tactile integration into robot learning for manipulation

There are several studies that integrate tactile sensors for manipulation with robot learning [[tactile-dp, 2025], [VT-WAM, 2026]].

26.07.03

Wear and tear might also be a problem of tactile sensors. The data distribution of a tactile used during training may differ from that used over time.

In reality reinforcement learning is impractical as (1) hard to get supervision on gentleness (2) object not reusable once damaged

Another point we can argue is that, common teleop system does not have force feedback, the demonstrator has to look at either camera feed or the arm to judge if the grasp is stable, it is extremely hard to get a correct grasp. While ours, since is finetuned by DPPO, is less sensitive to demo quality.

A reviewer might ask: why not active compliance control / impedance control on the parallel jaw. -> Actually we can try it out in sim.

We can try to reproduce stress-minimized quality metric.

One-shot/few-shot adaptation to deal with out-of-domain. Like the crazy drone paper in RSS26. Maybe even necessary for closing sim2real gap.

2026.07.21:

Summarizing recent experiments:

1. Without a stress penalty, the success rate goes up significantly, but the policy gets aggressive too.
2. With the stress penalty, we can achieve slightly improved success rate (0.96 sure better, aggregated with 3 random seeds) with slightly worse gentleness (rello), or a slightly less success rate with much better gentleness (domqg)
3. The training variance exists (rello, zrvpm, and avwoe)
4. Removing gradient clipping makes a policy worse in terms of success rate (0.99 percent sure by A/B test), see (trsmn v.s. rello), though gentleness goes up.
5. In summary, DPPO at the moment could be too hard to work

Some thoughts about next step:

1. We need some ways to compare policies in sim, taking gentleness into account. A/B test is ok, but needs to extend to stress-based metrics.
    a. Violin plot maybe useful
    b. Also, is it fair to summarize stress score across objects with different deformation parameters?
2. Shall we update the sim setup (like DR range) to make the task more difficult?
3. Shall we consider IsaacSim?

    a. Sure, but need to benchmark the throughput
4. We may try works that extend DPPO, like DSRL and/or LPS
    a. [Update]: LPS is not suitable here, it is offline RL
5. Consider the base policy architecture
    a. [Update]: checked, is the same as DP3 and MoDex (except for the action head), I'd say keep using the current architecture
6. Use privileged information in critics
7. Test the hypothesis that good sim policy naturally performs good in real life. If this doesn't hold, we need to pause and rethink.
    a. 3000 epochs may not be enough, might need 8000 epochs as DPPO did on robomimic tasks.
        i. [Update]: 8000 epochs indeed perform better, but only slightly, now is 0.77 success rate, the main challenge is finding a suitable grasp width
8. Teacher-student
9. Use Gaussian Splatting to reconstruct objects and put them in sim.
10. Train a real-world DP3 on mushrooms.
11. Evaluate with tactile.
12. Organize the code, make clear on how to run things
13. We might need to take object orientation into account, which makes demo collection via teleop extra harder.
    a. This might be necessary, imagine a tofu put in a different orientation.
    b. Using RL to make a policy realize this is too difficult, the base policy should already be able to do this
    c. This might be beneficial for us since DP3 seems to be poor at orientation generalization.

At the moment seems the BC policy with 300 demos do not perform that well, one potential reason is that the demo had too diverse initial pose, while in eval we only concern a narrow initial position. [Update] Not necessary is the demo, because as the plot shows there are still many demos lying at the center.

Recovery behavior did not appear tho, I think it is because when it failed the gripper opening is different from the initial conditions in demo.

As /home/kei/kei/gentle_manip/dataset/demos/mushroom_soft_batched/26-07-19-021443/inspect/maxy_ep259.mp4 show, the object point cloud becomes sparse as the arm moves down. Maybe we have to reduce the density of arm point cloud, this should be doable since we have the CAD.

By looking at the eval of ckpt 3000 of (gllzd), we confirm that the failures are purely from grasp failure, not approach failure. Approaches are all correct, it is just the grasp widths are wrong sometimes.
(single_lift_mushroom_soft_pcd_wide/gllzd/eval/2026-07-21_18-33-15)

This suggests that with grasps that are so delicate, a vision-only imitation learning policy has a hard time performing well against so many variations including pose, size, shape, etc. Trying to cover many with demo is impractical, and a base method can't generalize well.

A possible idea is then train a policy that has stress information (which gives signal for appropriate gripper command), and distill it with many many demos (potentially over 1k)

My hypothesis is that if the demonstrations are more aggressive, the pick policy would result in almost 100% success, just like those in robomimic.

It might be good to keep the initial scene observation to deal with the object being occluded when manipulation

OmniReset uses 8k trajectories for distillation, that is a lot.

Tested xxiaw-8000 in real, for the first time on mushrooms. The grasp success rate is 17/33 (0.515), safe grasp success rate is 4/33 (0.121) (i.e. 4/17 was safe). There is still room for improvement. The next step is to evaluate the real-world DP under the same protocol. One concern is that despite the fact that we evaluate on the same set of positions, we can't guarantee the same size/shape/orientation is used across the same positions in different runs, how much does this affect the result?

With 1k demos, it is still difficult to get a high grasping performance, the failure mode is still missed due to inappropriate grasping width. This may suggest that coverage can't solve the problem. The problem may lie somewhere else. E.g. object pnts too sparse as approach, or pnt cld is just too ambiguous for precise grasp. Though the demos contain initial positions that are not used during eval, not sure if this would affect the conclusion. Should we check?

Confirm that the demo is generating different grasp widths, which is good. See /home/kei/kei/gentle_manip/dataset/demos/mushroom_soft_batched/26-07-19-021443/inspect/maxx_ep80.mp4

Running train on 150 and 300 (subsampled from 1k) demos, its purpose is 2. (1) check with validation loss we just introduced (2) see if data size changes anything (compare with 1k)

An observation is that as the gripper gets closer to the object, more gripper points are introduced and the object points become sparse (see /home/kei/kei/gentle_manip/dataset/demos/mushroom_soft_batched/26-07-22-005407/inspect/pcd_ep0940.mp4)

Current status as of 2026.07.23:

We found that the number of demonstrations does not affect the task performance significantly (Compare keqkk with 1k demos, and xxiaw with 300 demos, and 2026-07-04_20-03-36_42 with 50 demos, they all get about 0.7-0.75). The failure mode is always wrong grasping width. There are few potential explanations:

1. Point cloud is just not enough to infer correct grasping width
2. When the gripper approaches the object, object point cloud becomes sparse, not sufficient for policies to decide the correct grasping width
3. Network architecture / hyper-parameter problem
4. The size variation is just too little, with 1k episodes, there are only 40 different shapes.

To test for hypothesis 2, I am letting the point cloud to focus near the object and gripper. Note that this is a dirty approach and wouldn't generalize. We are only doing this to verify if more object points could improve the policy performance.

If hypothesis 2 is ruled out, we can test for hypothesis 1 by replacing point cloud with object center location and stress value. Basically a privileged state-based policy.

To test hypothesis 4, we can use size variation frequency of 1 or 2 batches to collect more diverse cases. Training on those demo will directly tell us if there is any difference.

Btw, as data collection on 5 envs show, the FPS is overall ~19 (~4 per env)

```
[Genesis] [13:57:14] [INFO] Running at 19.46 FPS (3.89 FPS per env, 5 envs).
[Genesis] [13:57:14] [INFO] Running at 19.48 FPS (3.90 FPS per env, 5 envs).
[Genesis] [13:57:15] [INFO] Running at 19.49 FPS (3.90 FPS per env, 5 envs).
```

Other considerations:

1. per 4 batch shape randomization may not be enough for a thorough eval
2. 100 episodes eval may not be so stable, maybe need 150 or 200 episodes
    a. But it is indeed slow. In this regard IsaacSim might be better, if it works

gpieh has validation loss skyrocket after 1.5k, let's see how later ckpts perform.

Now things are getting a bit messy. Is it possible to design some sort of html page to display the experiment results? Maybe we can have a script to generate the page as a html file, which can be opened in browsers. I am not sure how the page should look like. I guess I will list some wishes that the page could feature: (1)

A good policy to handle fragile food should satisfy the following aspects:

1. Can handle variations in position, orientation, size, shape, and etc. (maybe friction, stiffness, fragility)
2. Should have high success rate
3. Safe
4. Should be robust to accidents like slippage, rolling, etc. and induce retry behavior

Naive real-world BC can handle variations in shape, but hardly orientation and other properties. Also it is hard for human tele-operators to produce high quality demos given that only vision can be relied on. Plus it is labor intensive.

A BC policy trained with many simulation samples generated by naive scripted grasping expert can have a better success rate but still induce a high damage rate. Plus it can't handle object orientation variation.

A DPPO finetuned BC policy (trained on naive scripted grasping) can have higher success rate but still damage the food significantly. Still can't handle object orientation variation.

The observation that raising the number of demo doesn't improve the performance much is aligned with what DemoGen paper observes in their appendix.

Also the focus_object line of experiment currently shows severely degraded performance, not sure what went wrong. I feel this is sooner or latter necessary because we need the policy to see the object clearly to decide what grasp pose to use.

Also updated the rotation action scales (in the meantime reduced linear motion scales) to better accommodate grasp pose varying cases.

The tactile DP3 works (on red cube), though not perfect and we did not measure rigorously, but at least it is non-degenerative.

We could consider using a better rotation representation than quaternion, but not sure if that is necessary.

Another way to frame it is to say that real world food come in various orientation, size and position, etc. Mere 50-100 demos are not likely to be enough, and may suffer sever covariate shift problem. So we propose to use sim2real for coverage. This requires testing various previous methods.